DV300 Project Proposal

MonoFact

Learn the truth, one swipe at a time

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Interactive Development 300

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MonoFact

"Learn the truth, one swipe at a time."

App Description

MonoFact is an educational cross-platform mobile application that helps users distinguish between facts and myths through an engaging swipe-based game. Users swipe right if they believe a statement is a fact and left if they believe it is a myth. After every swipe, the app provides immediate feedback and a concise explanation, encouraging users to learn while they play.

Conceptualisation & Problem Statement

The inspiration for MonoFact came from combining educational content with intuitive interactions. Rather than presenting users with a long article or traditional quizzes, the application uses swipe gestures to create a more engaging and accessible experience. The concept was influenced by the increasing spread of misinformation online and the popularity of modern mobile interfaces that encourages quick, interactive engagement.

CARDS:

Learning something, Only one colour, and swiping

Problem Statement

Misinformation has become increasingly common across social media and online platforms. Many users struggle to identify whether information is accurate because educational resources are often lengthy or unengaging.

MonoFact addresses this issue by presenting users with short statements that they classify as facts or myths. Each answer is immediately followed by an explanation, allowing users to continuously improve their knowledge through active participation.

Target Audience

MonoFact is designed for:

- Students
- Teenagers
- Trivia enthusiasts
- Casual learners
- Anyone interested in improving their general knowledge

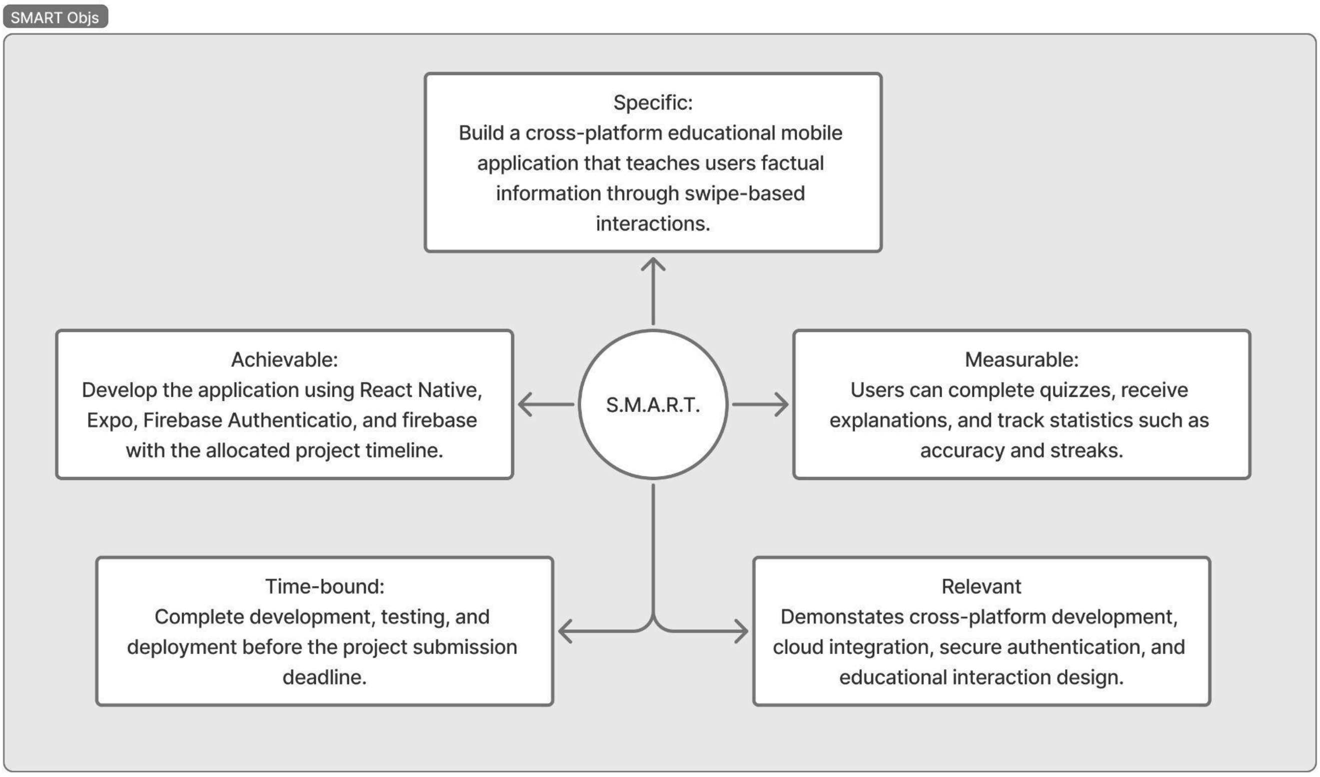
The application is intended for users who enjoy learning in short sessions commuting, taking breaks, or relaxing.

Core Features

Project Scope

The application focuses on creating an engaging educational experience while demonstrating the use of modern cross-platform mobile development technologies.

SMART Objective



Feature Prioritisation

MVP (Must Have)

- User registration
- User Login
- Firebase Authentication
- Category Selection
- Swipe left (Myth)
- Swipe Right (Fact)
- Educational Explanations
- User Statistics
- Firestore Database
- Android & iOS Support

Nice to Have

- Daily Challenge
- Favourite Facts
- Achievements Badges
- Difficulty Levels
- Search Categories

Future Considerations

- Multiplayer
- Leaderboards
- Admin dashboard
- AI-generated Fact Suggestions
- Push Notifications
- Offline Support

User Roles (Admin role will not be included during the initial release)

1. Standard User
    - Create an account
    - Log in securely
    - Choose categories
    - Play quizzes
    - View statistics
    - Manage their profile

User Stories

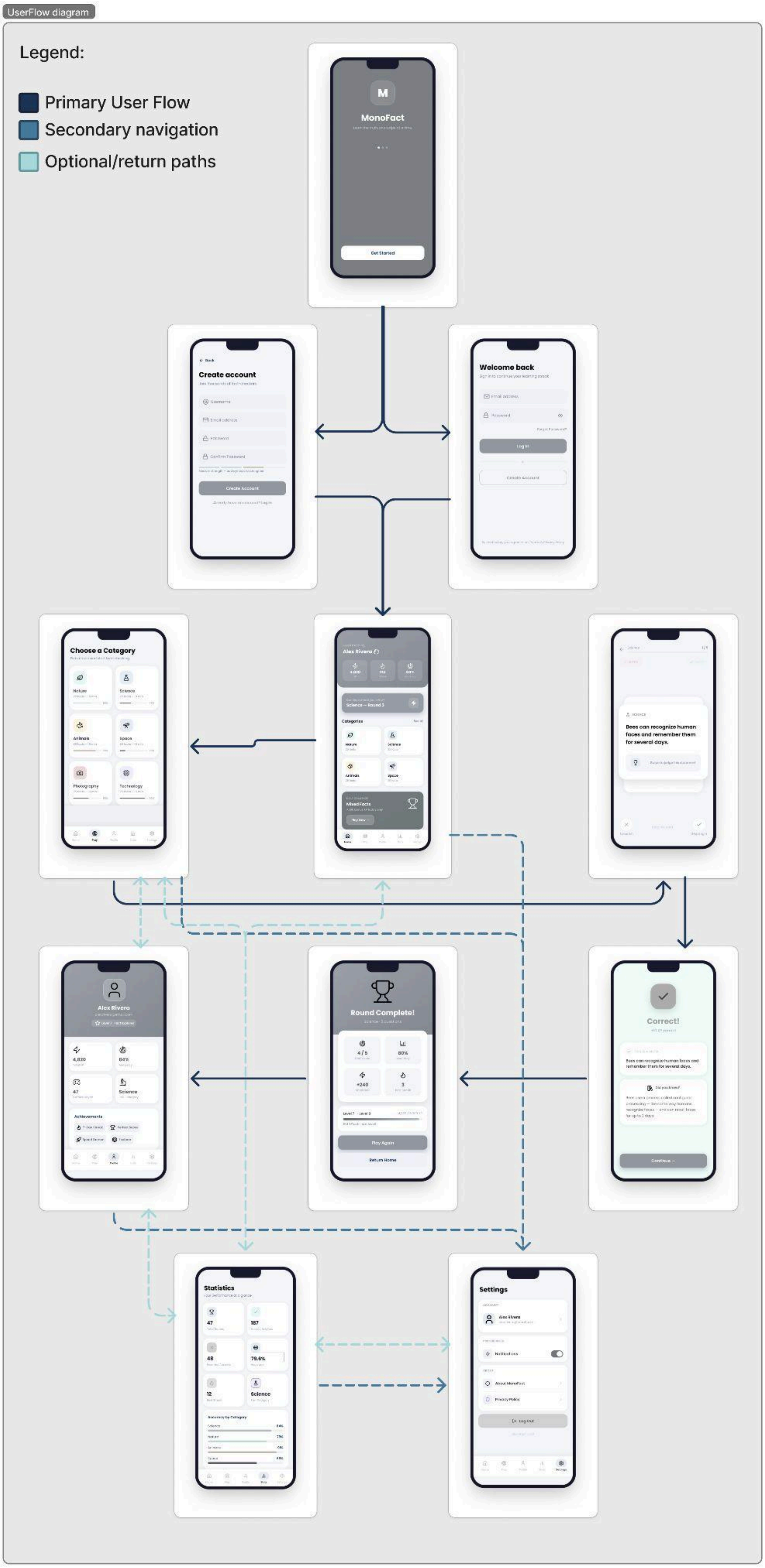
As a new user, I want to register an account so that my progress can be saved.

As a returning user, I want to log in so that I can continue learning.

As a learner, I want to select different categories so that I can understand why my answer was correct or incorrect.

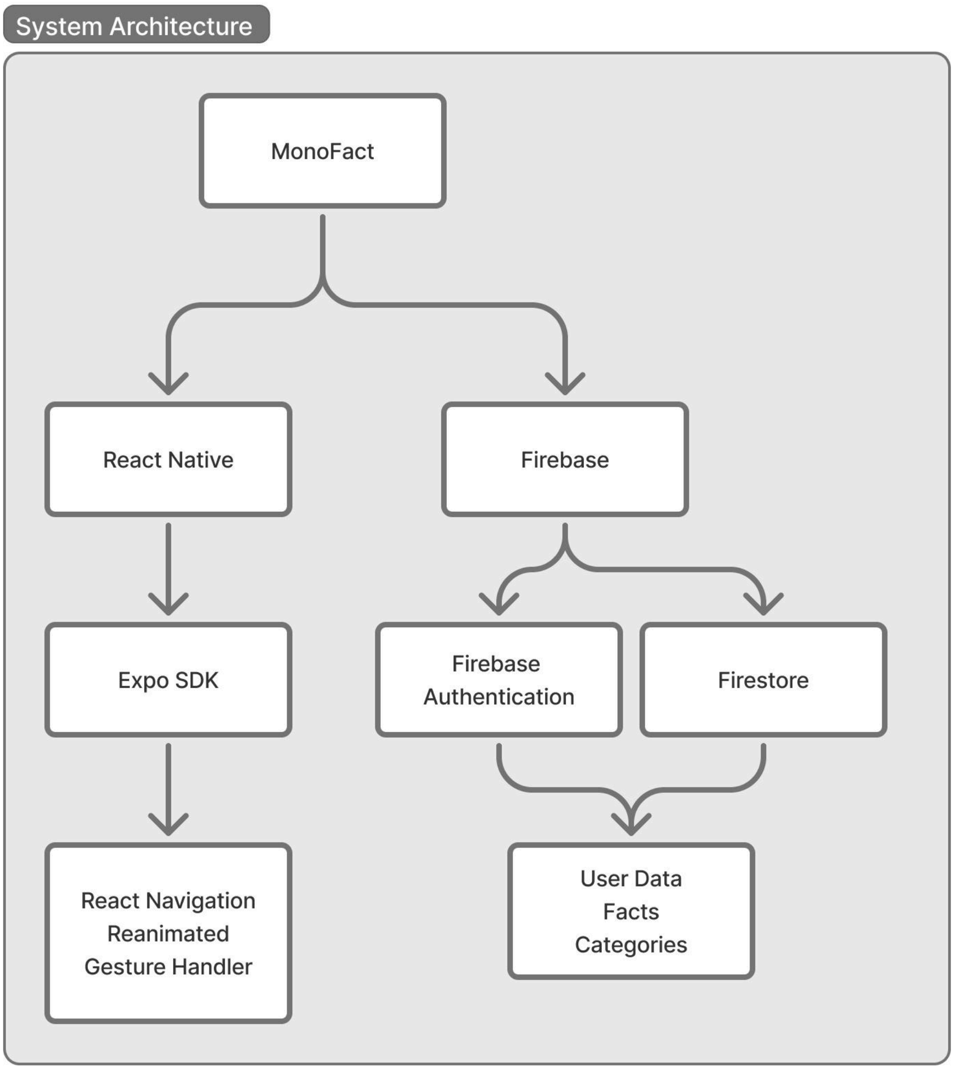
As a user, I want to view my statistics so that I can measure my improvement over time.

User Flow


UserFlow diagram
Legend:
Primary User Flow
Secondary navigation
Optional/return paths
M
MonoFact
Create account
Welcome back
Choose a Category
Alex Rivera
Correct!
Round Complete!
Statistics
Settings

Technical Planning

Technology	Purpose
React Native	Cross-platform mobile development
Expo	Simplifies development and deployment
TypeScript	Improves code quality and maintainability
Firebase Authentication	User authentication
Cloud Firestore	Cloud-hosted NoSQL database
React Navigation	Screen navigation
React Native reanimated	Swipe animations
React Native Gesture Handler	Gesture recognition

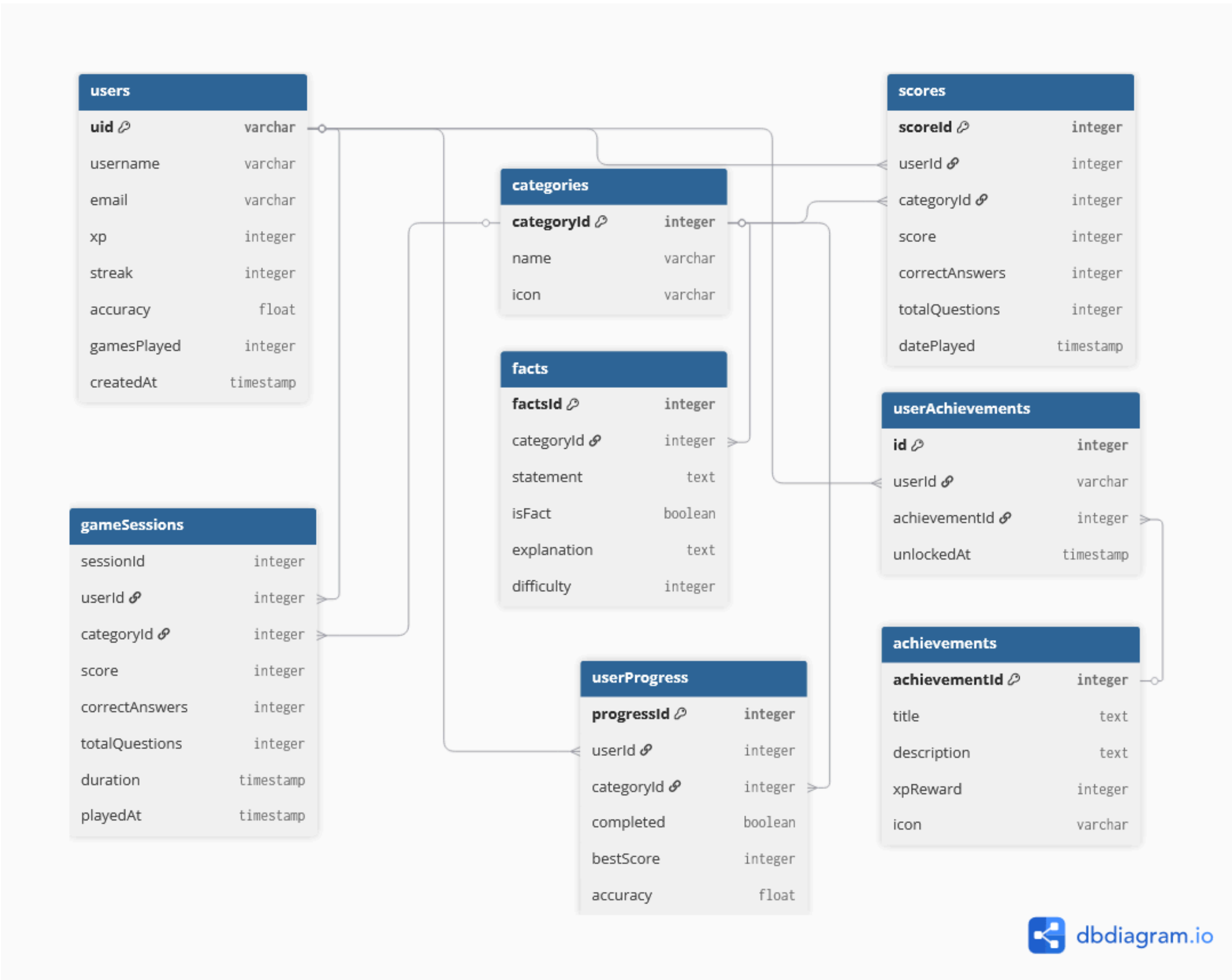


Technical Challenges

Potential challenges include:

- Implementing Smooth swipe animations across Android and iOS.
- Designing an efficient Firestore database structure.
- Managing user authentication securely.
- Maintaining responsive performance on different devices.
- Tracking user statistics in real time.

Firestore Database



Wireframes & UI/UX Considerations

Design Inspiration

MonoFact will follow a clean, modern design inspired by educational and interactive applications such as:

- Duolingo
- Tinder (card interactions only)
- Headspace
- Material Design
-

Typography:

Poppins

UI Components:

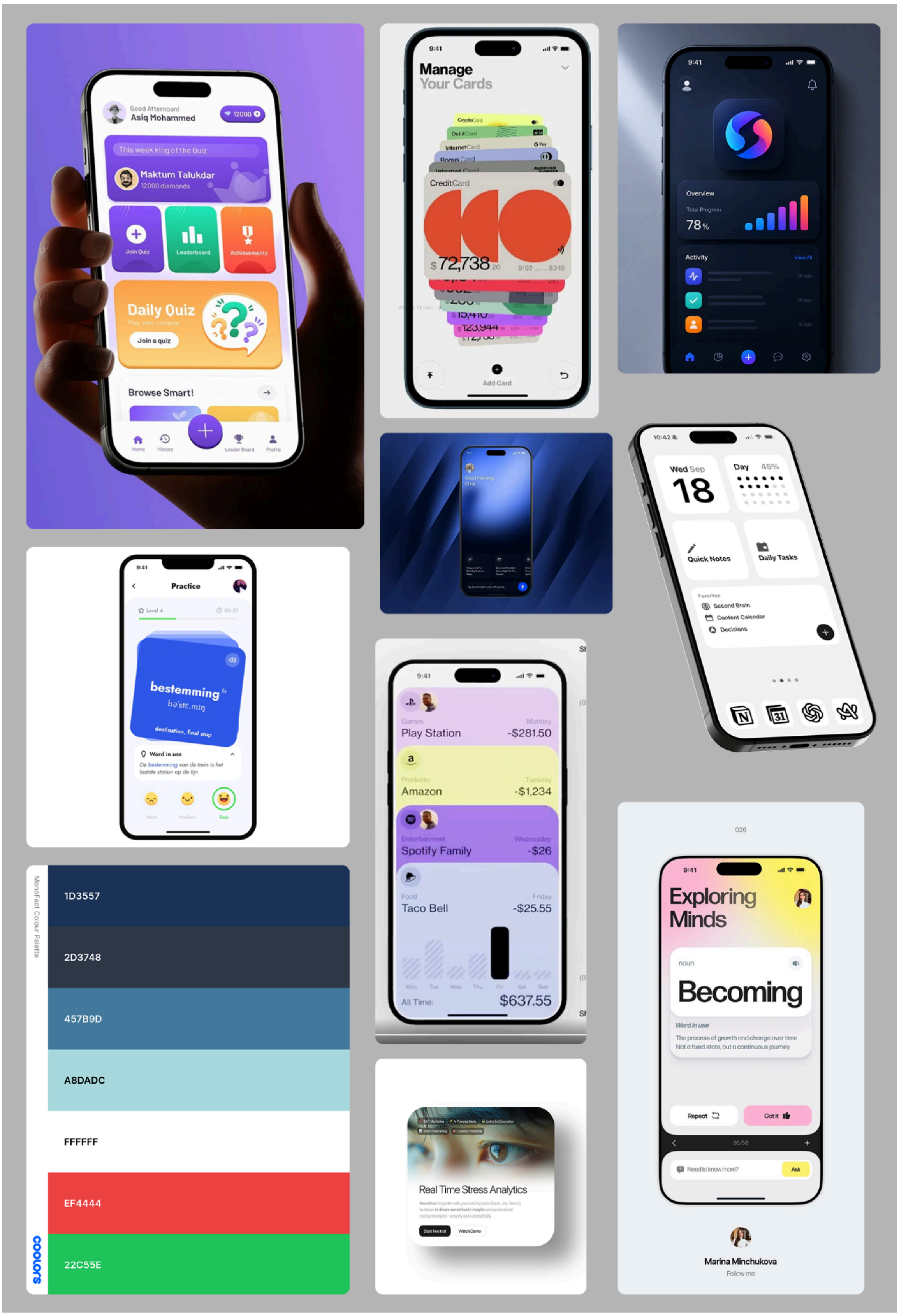
- Rounded cards
- Rounded buttons
- Card Swipe Animation
- Minimal Icons
- Large Readable Text
- Consistent Spacing

Accessibility:

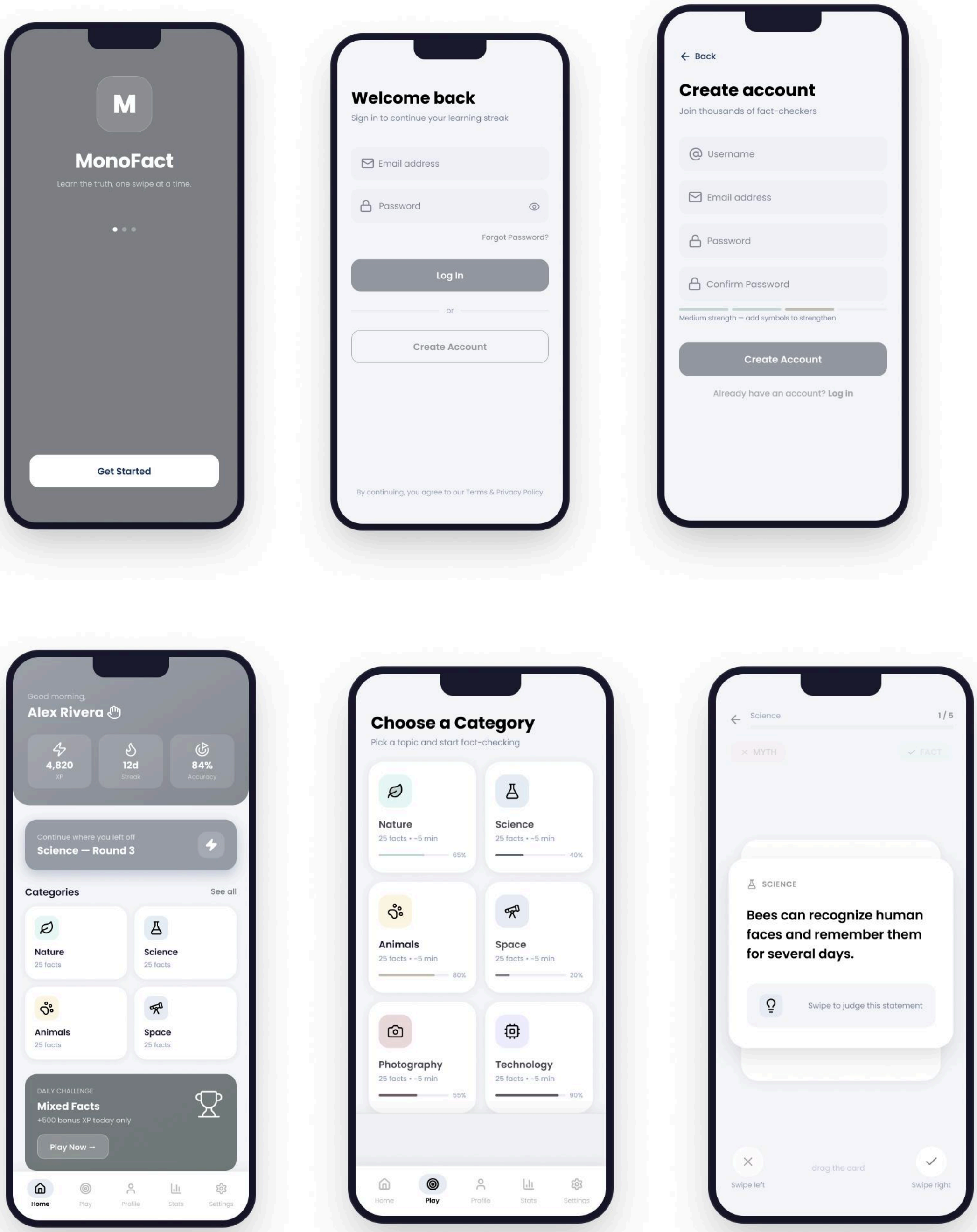
The application will:

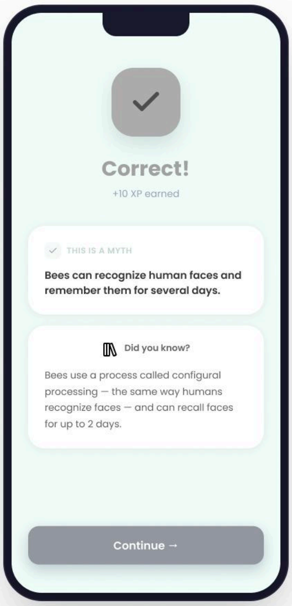
- Use high- contrasts colours
- Maintain readable typography
- Provide large touch targets
- Give Immediate visual feedback
- Ensure consistent navigation across all screens

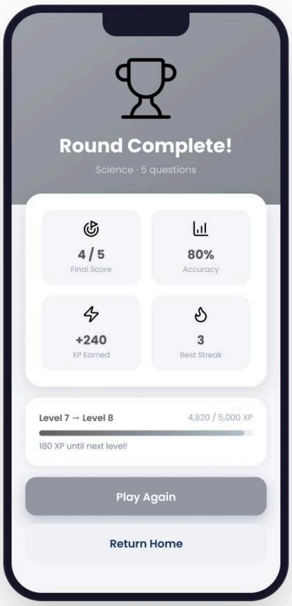
Moodboard & Colour Palette:

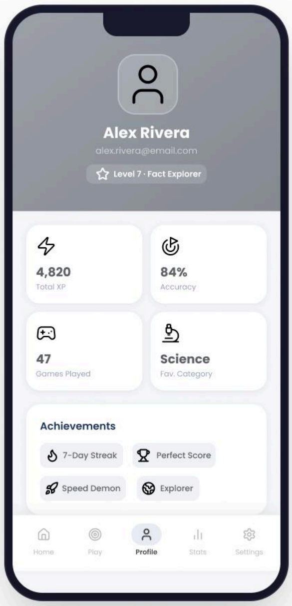


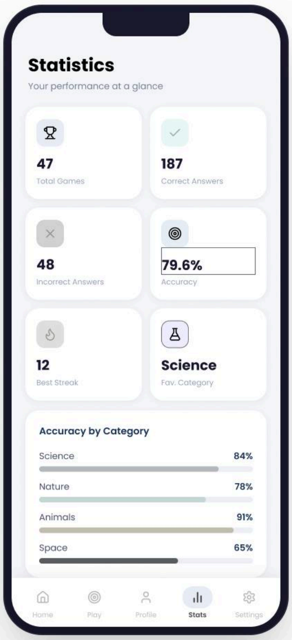
Wireframes: Final will follow the proposed navy-blue design system.

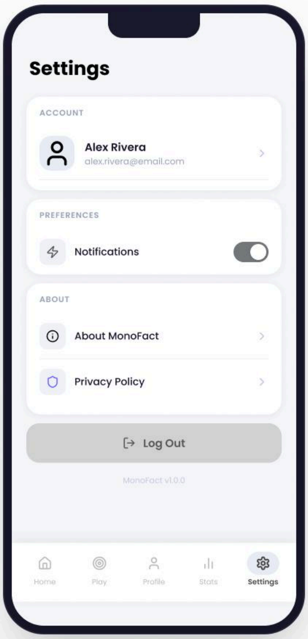



Correct!
+10 XP earned
THIS IS A MYTH
Bees can recognize human faces and remember them for several days.
Did you know?
Bees use a process called configural processing — the same way humans recognize faces — and can recall faces for up to 2 days.
Continue →


Round Complete!
Science · 5 questions
4 / 5
Final Score
80%
Accuracy
+240
XP Earned
3
Best Streak
Level 7 → Level 8
4,820 / 5,000 XP
180 XP until next level!
Play Again
Return Home


Alex Rivera
alex.rivera@email.com
Level 7 · Fact Explorer
4,820
Total XP
84%
Accuracy
47
Games Played
Science
Fav. Category
Achievements
7-Day Streak
Perfect Score
Speed Demon
Explorer
Home
Play
Profile
Stats
Settings


Statistics
Your performance at a glance
47
Total Games
187
Correct Answers
48
Incorrect Answers
79.6%
Accuracy
12
Best Streak
Science
Fav. Category
Accuracy by Category
Science
84%
Nature
78%
Animals
91%
Space
65%
Home
Play
Profile
Stats
Settings


Settings
ACCOUNT
Alex Rivera
alex.rivera@email.com
PREFERENCES
Notifications
ABOUT
About MonoFact
Privacy Policy
Log Out
MonoFact v1.0.0
Home
Play
Profile
Stats
Settings

Project roadmap

Week / Data	**Tasks**
Week 1 (6 - 12 July)	• Brainstorm app concept based on inspiration cards • Research similar applications (Duolingo, Tinder, Headspace) • Define project scope
Week 2 (13 - 19 july)	• Complete project proposal • Design wireframes in Figma • Create ERD and user flow • Plan Firestore collections
Week 3 (20 - 26 July)	• Initialise GitHub repository • Create Expo React Native project • Configure Firebase Authentication and Firestore • Build Login & Register screens
Week 4 (27 Jul - 2 Aug)	• Develop Home screen • Implement category selection • Build swipe card gameplay • Add animations and navigation
Week 5 (3 - 9 Aug)	• Connect gameplay to Firestore • Save game sessions and statistics • Develop Profile and Statistics screens
Week 6 (10 - 16 Aug)	• Implement achievements and XP system (if time allows) • Improve UI and accessibility • Test on Android and iOS devices • Fix bugs

Week 7 (17 - 19 Aug)	• Final testing • Optimise performance • Record demo video • Submit proposal, source code and application

Version Control

GitHub will be used throughout the development of the application

Branch stats:

- main
- development
- feature/auth
- feature/gameplay
- feature/profile
- feature/db

Conclusion

MonoFact combines interactive mobile design with educational content to create an engaging learning experience that encourages users to think critically about the information they encounter. By using React Native and Firebase, the project demonstrates modern cross-platform development practices while remaining achievable within the project timeframe. The application satisfies the assignment requirements by incorporating auth, cloud-based data storage, gesture-based interaction, and educational functionality into a polished and user-friendly mobile experience.